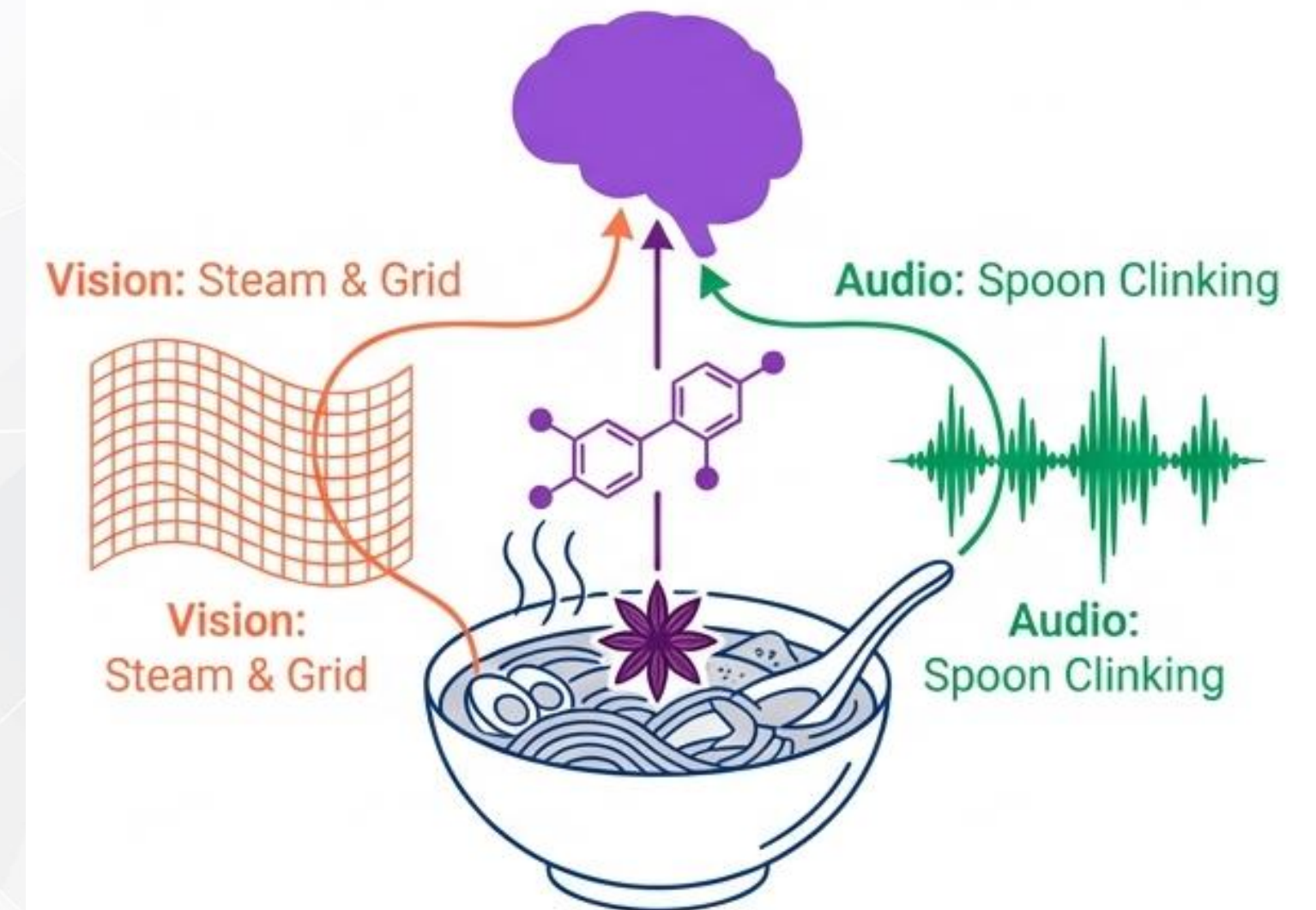
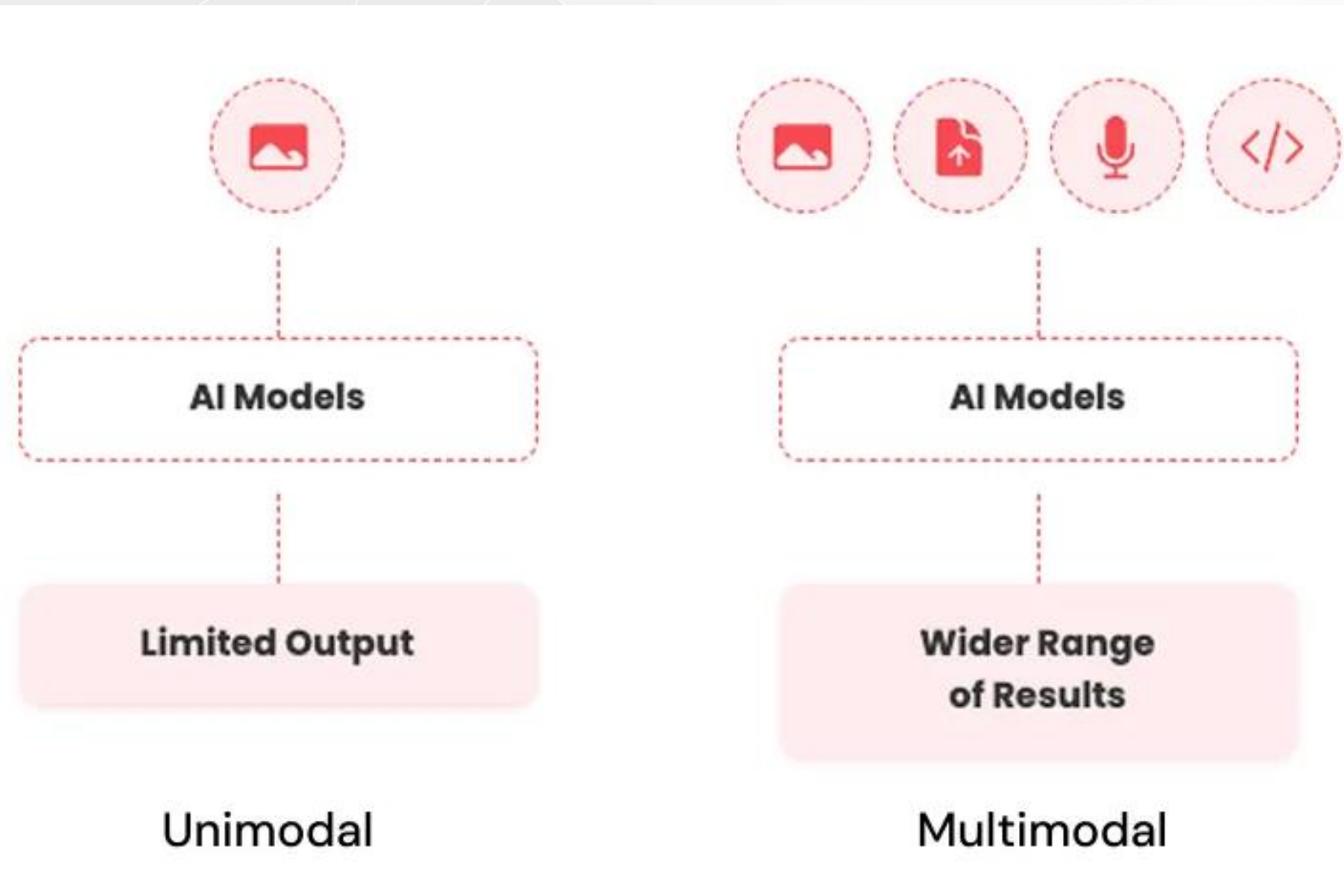




VN-UK
Institute for Research
& Executive Education

Multimodal Machine Learning

Unimodal vs Multimodal



Unimodal vs Multimodal



Unimodal

Limited to a single data type.
Vulnerable to noise or missing
information within that specific
channel.



Multimodal

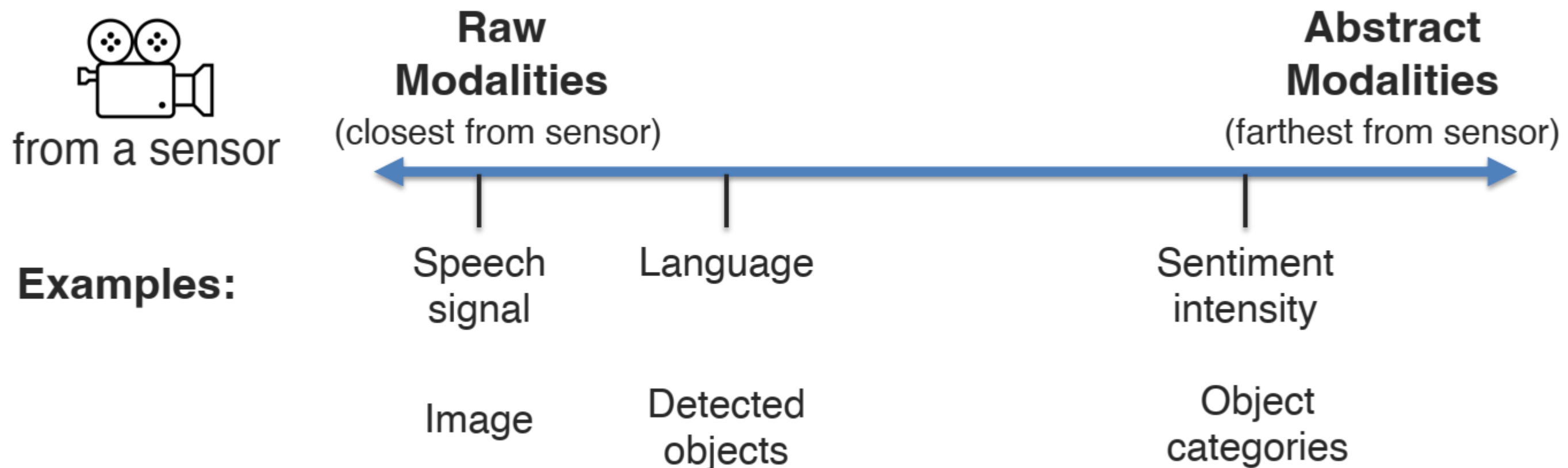
Captures richer context. More
robust; can leverage
secondary modalities when the
primary is compromised.

What is a Modality?

Modality: the way information is expressed or perceived, e.g., language, vision, sound, touch.

From a sensor: raw modality is closest to the physical signal, abstract modality is derived.

The same phenomenon can appear across multiple modalities (e.g., emotion in voice and face).



Multimodal Data



Video lecture

- frames (image)
- audio (speech)
- transcript (text)
- slides (graphics)



Patient record

- MRI images
- ECG time series
- lab tests
- doctor narrative notes

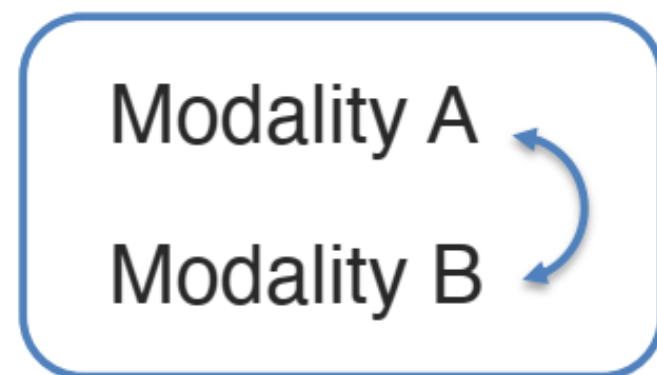


Navigation

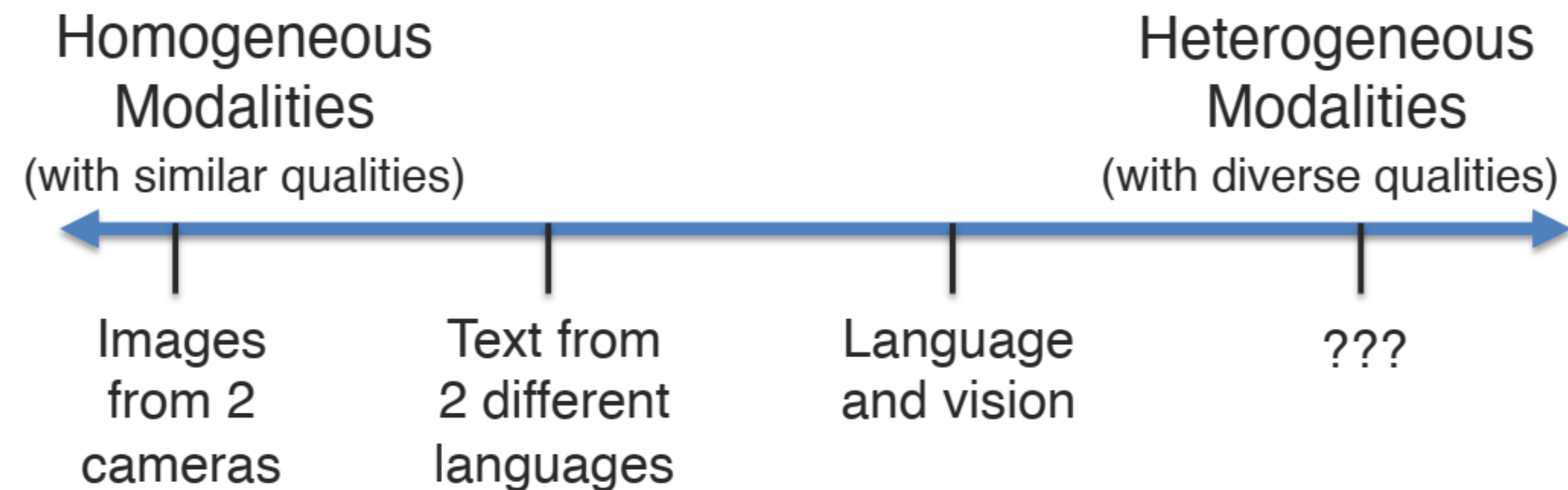
- street images
- GPS traces
- map graphs
- IMU sensors

Heterogeneous Modalities

Information present in different modalities will often show diverse qualities, structures and representations.



Examples:





Dimensions of Heterogeneity

1. Structure

Static, temporal, spatial, or hierarchical.

2. Repr. Space

Discrete tokens vs. Continuous embeddings.

3. Information

Entropy, density, and information overlap.

4. Granularity

Mismatched sampling rates and resolutions.

5. Noise

Uncertainty and modality-specific noise.

6. Relevance

Task-specific importance of each mode.



Structure

Static: single images, fixed embeddings.

Temporal: speech, videos, sequential sensor data.

Spatial: 2D images, 3D point clouds, spatial grids.

Hierarchical: document $\rightarrow$ sentence $\rightarrow$ word, scene $\rightarrow$ object $\rightarrow$ part.



Representation Space

Discrete: tokens, symbolic labels, graph nodes.

Continuous: embedding vectors, feature maps, latent states.

Interpretable vs. latent features (explicit scores vs. black-box features).



Information

Entropy: how uncertain or variable a modality is.

Density: information per unit (per frame, per second, per token).

Overlap and range: shared vs. unique information, and what each modality can represent.



Granularity

Different sampling rates: audio (kHz), video (fps), text (tokens/s).

Resolution: high-res images vs. coarse sensors; precision levels.

Granularity mismatch complicates temporal/spatial alignment and fusion.

Noise



Each modality has its own noise distribution and SNR.

Visual Noise

Motion blur, occlusion, variable lighting, sensor artifacts.

Acoustic Noise

Background chatter, wind noise, ASR transcription errors.

Packet Loss

Dropped sensor packets or latency in real-time streams.



Relevance

Not all modalities are equally useful for every task → The importance of a modality is context-dependent.

Emotion Detection: Prosody and facial expressions dominate.

Night Driving: Lidar/Radar are more relevant than RGB cameras.

→ Dynamic Weighting: Modern models use gating or attention mechanisms to focus on the most relevant modality at any given moment.



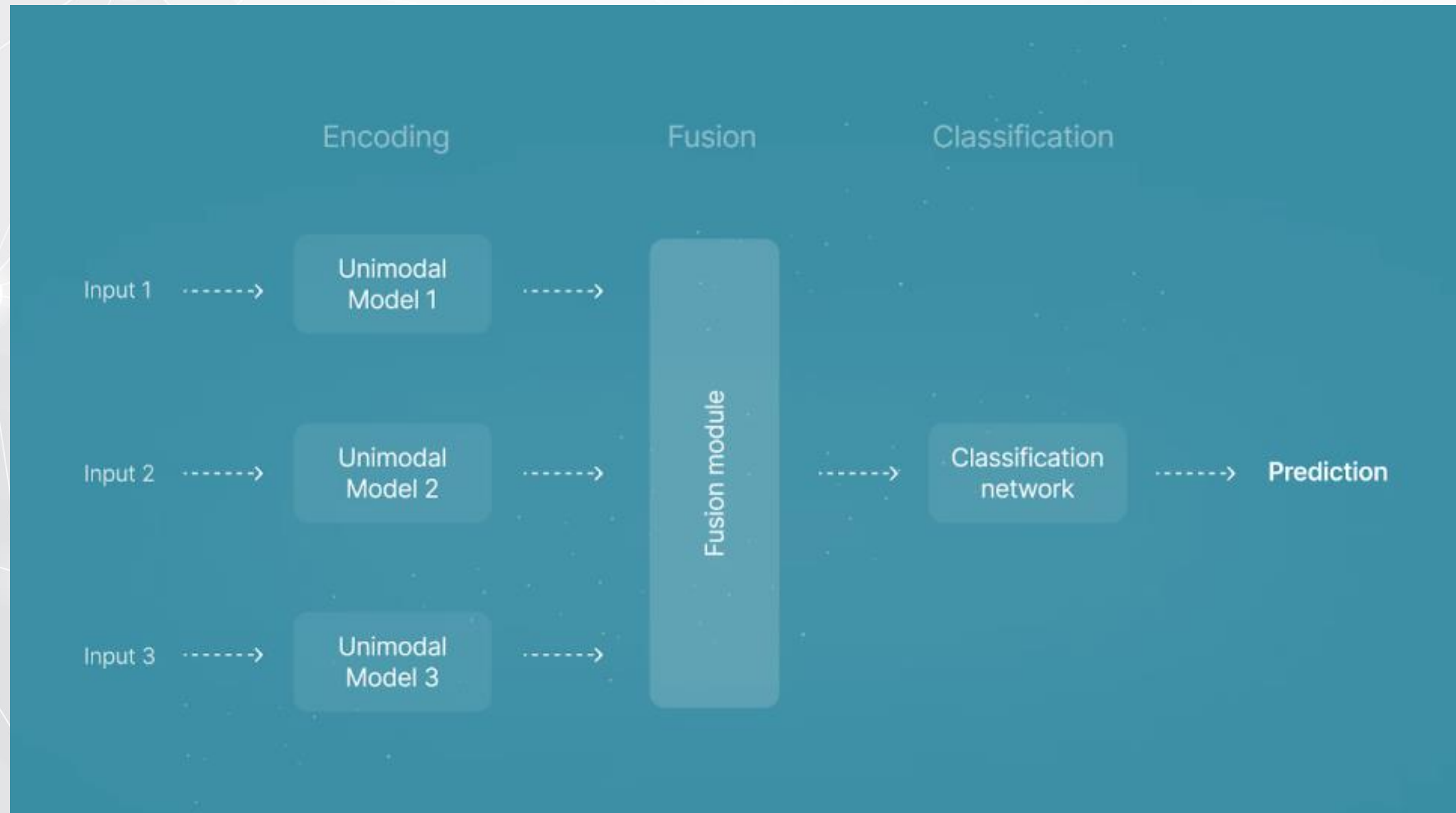
Cross-Modal Interactions

Redundancy: multiple modalities carry the same signal (speech + gesture).

Dominance: one modality overrides another when they conflict.

Modulation and emergence: interpretation or new meaning depends on combining modalities.

Multimodal Pipeline



Multimodal Pipeline

Preprocess each modality separately (normalization, tokenization, feature extraction).

Encode each modality into latent representations.

Fuse or interact representations, then apply task-specific heads for prediction or generation.



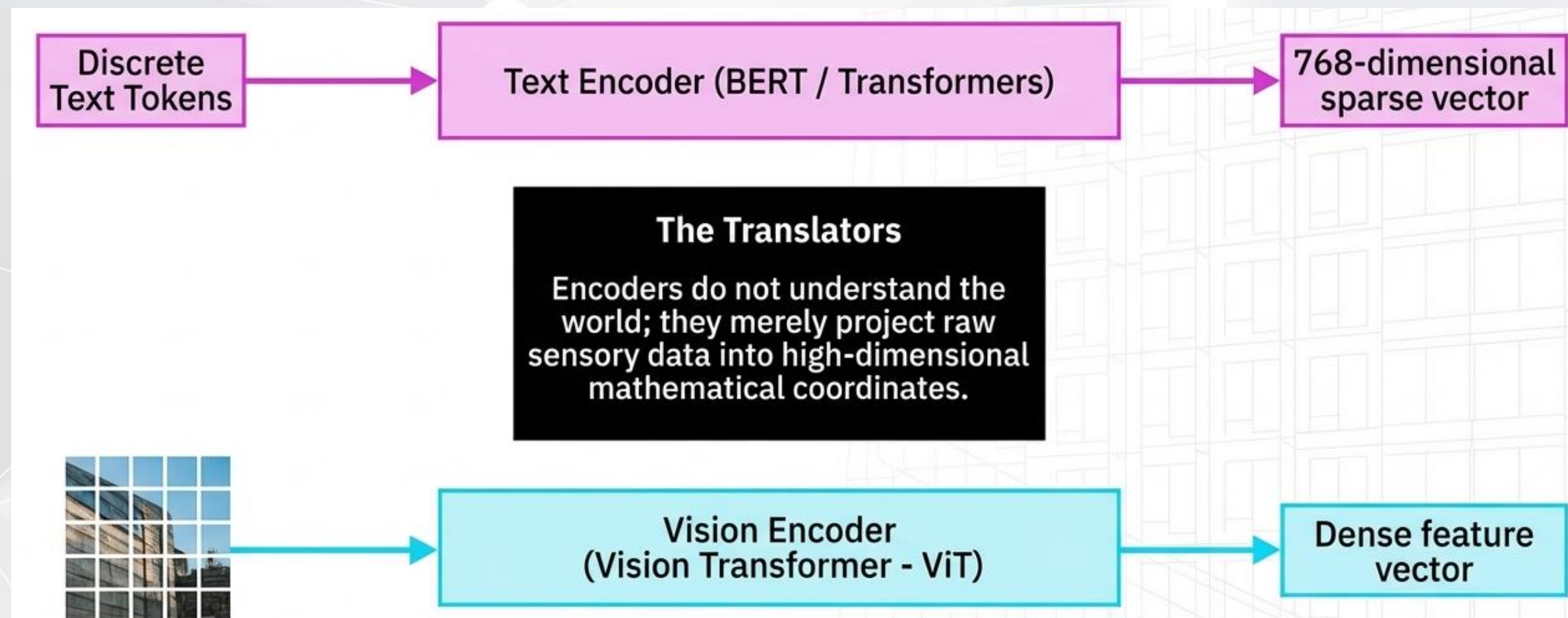
Input Module

Modality-Specific Encoders

Text: tokenization → embeddings → transformer encoder.

Image: CNN or Vision Transformer extracting spatial features.

Audio/sensors: temporal models over spectrograms or numerical features.





Fusion Module

The "Heart" of the system where individual data streams are merged.

Early Fusion

Concatenate raw or low-level features immediately.

Intermediate

Cross-modal attention between modality encoders.

Late Fusion

Combine independent unimodal predictions.

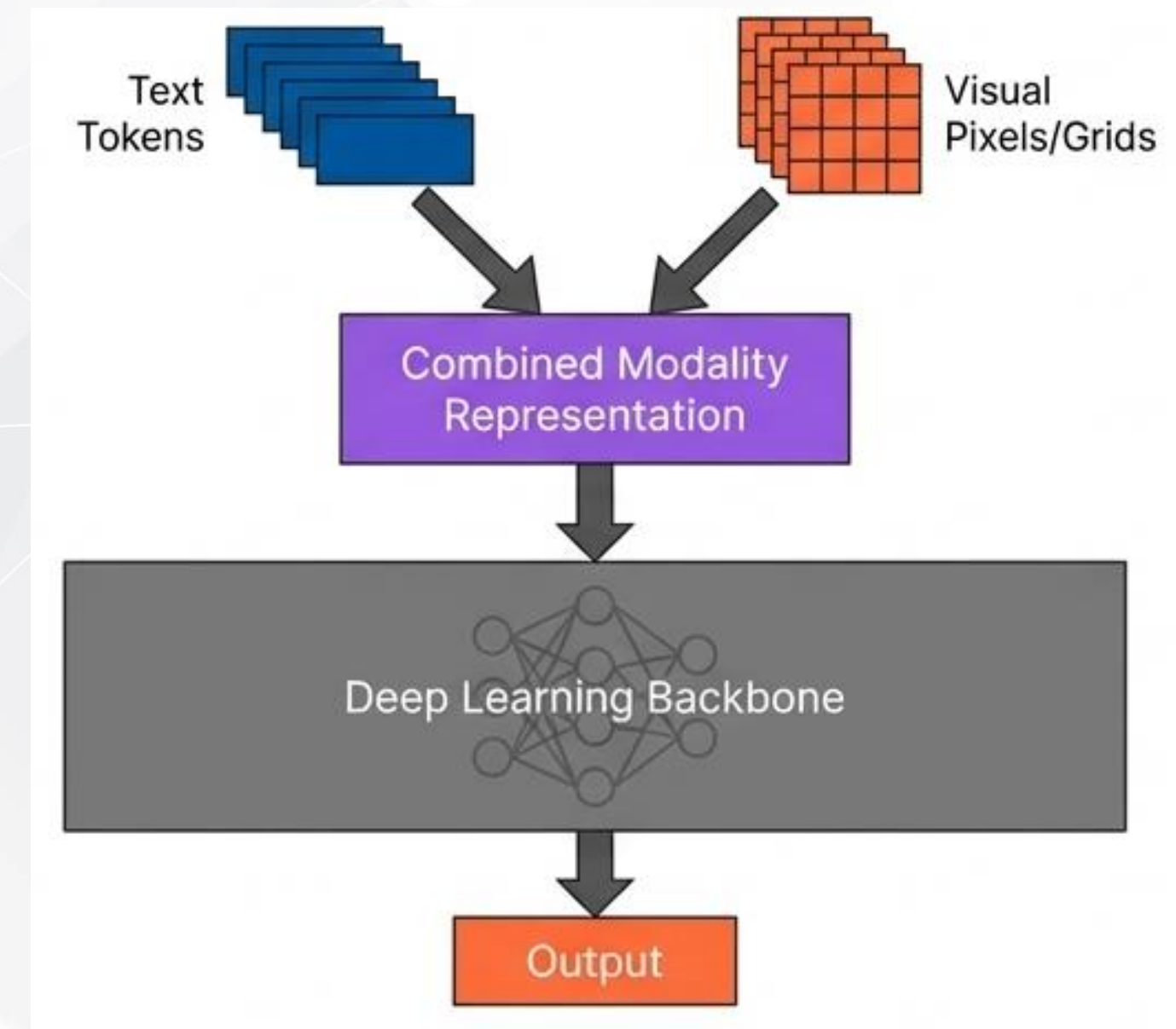
Early Fusion (Feature-level)

Mechanism: Raw features are combined at the beginning of the network.

Pros: Captures high-level interactions between modalities early on, simple and parameter-efficient

Cons: Can struggle with strong heterogeneity and misalignment (e.g. 60fps videos and 44kHz audio), dimensionality explosion, problem with dimensionality imbalance.

Example: Concatenating text embeddings with image pixels to describe a scene.

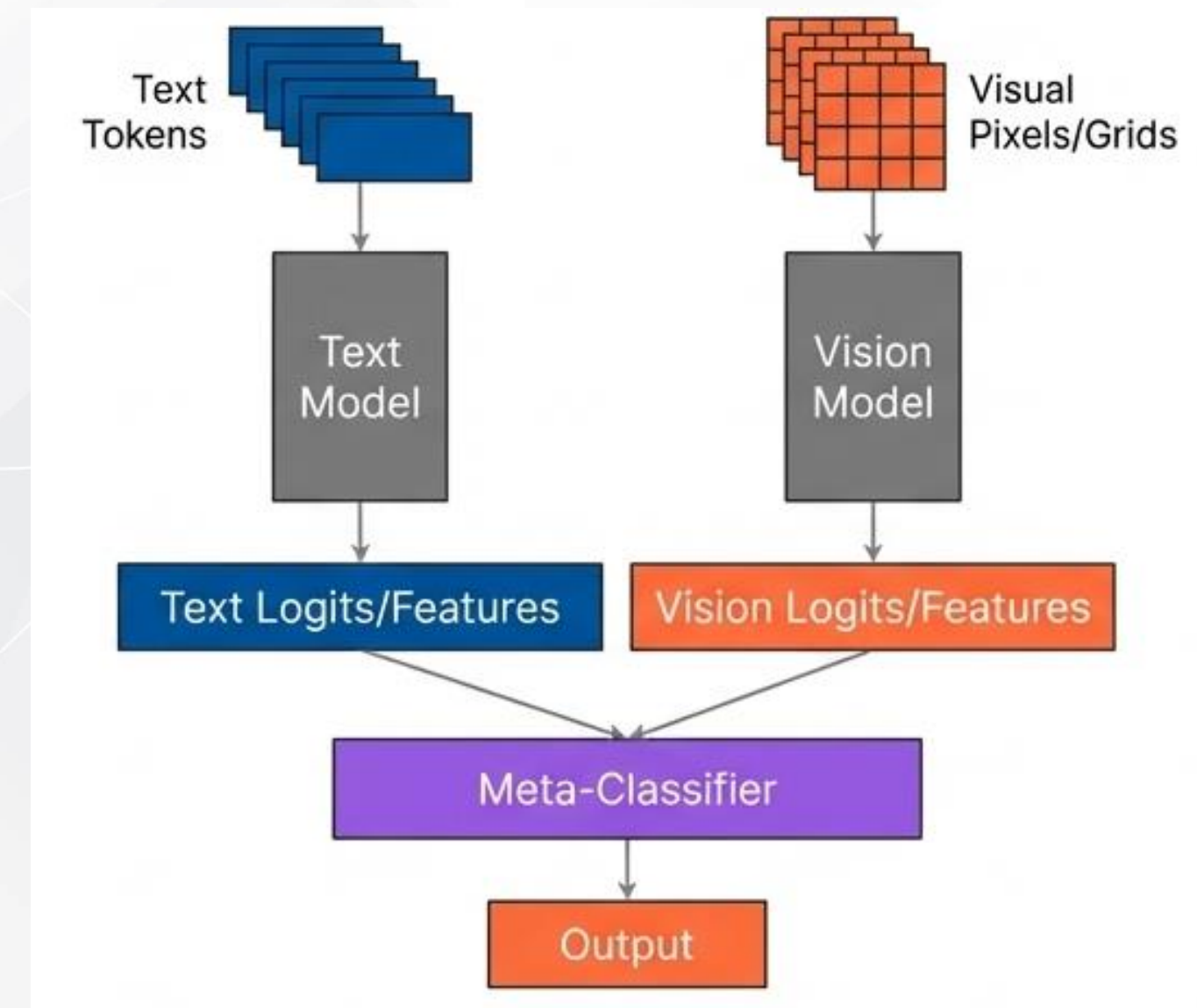


Late Fusion (Decision-level)

Mechanism: Each modality is processed by an independent model. Only the final scores/predictions are combined (e.g., via averaging or voting or using a meta-learner).

Pros: Robust; if one modality is noisy (e.g., a blurry image), the other (e.g., clear audio) can still carry the prediction.

Cons: Loss of correlation, cannot capture cross-modal interactions.



Intermediate Fusion (Feature-Level)

Intermediate fusion encodes modalities separately but merges their internal "hidden" representations (feature vectors) before the final decision-making layer.

Separate encoders; interaction via cross-attention or shared transformer blocks.

Supports fine-grained alignment (tokens $\leftrightarrow$ image patches, frames $\leftrightarrow$ sentences).

Common in modern vision–language models.

- Gated Multimodal Units
- Cross-Attention

Gated Multimodal Fusion

Uses a "gate" (similar to an LSTM or GRU) to decide how much influence each modality should have on the final feature representation based on the input context.

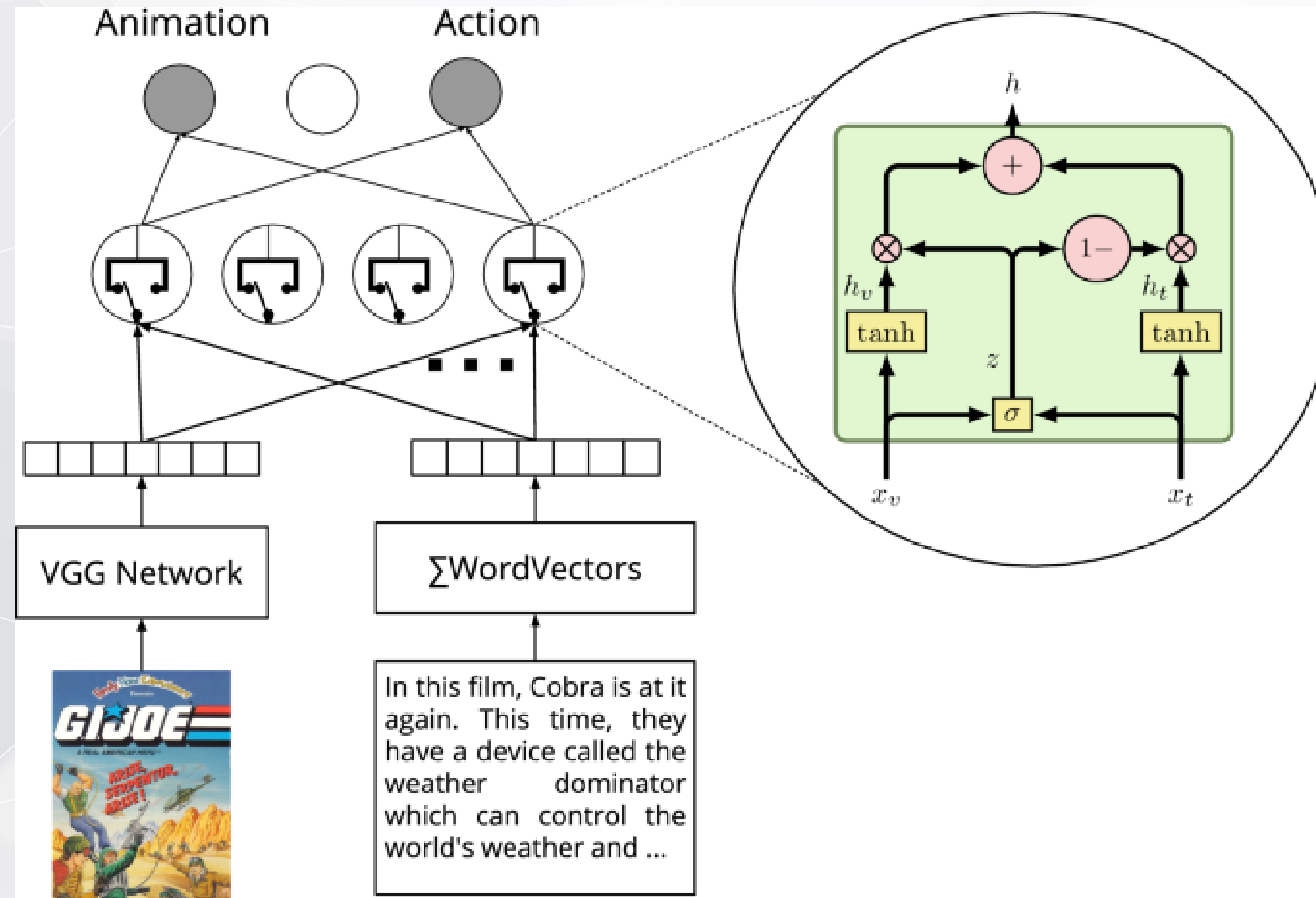
Pros:

- Dynamic Weighting: The model can decide to "listen" more to the audio if the image is blurry, or "look" more at the video if the audio is noisy.
- Semantic Alignment: Fusing at higher layers allows the model to connect abstract concepts rather than just raw pixels.

Cons:

- Architectural Complexity: Requires designing sophisticated interfaces between different types of neural networks.
- Parameter Heavy: Often leads to significantly larger models that require careful hyperparameter tuning and large-scale training.

Gated Multimodal Fusion



Cross-Attention

Instead of a model only looking at its own tokens (Self-Attention), one modality "queries" the other.

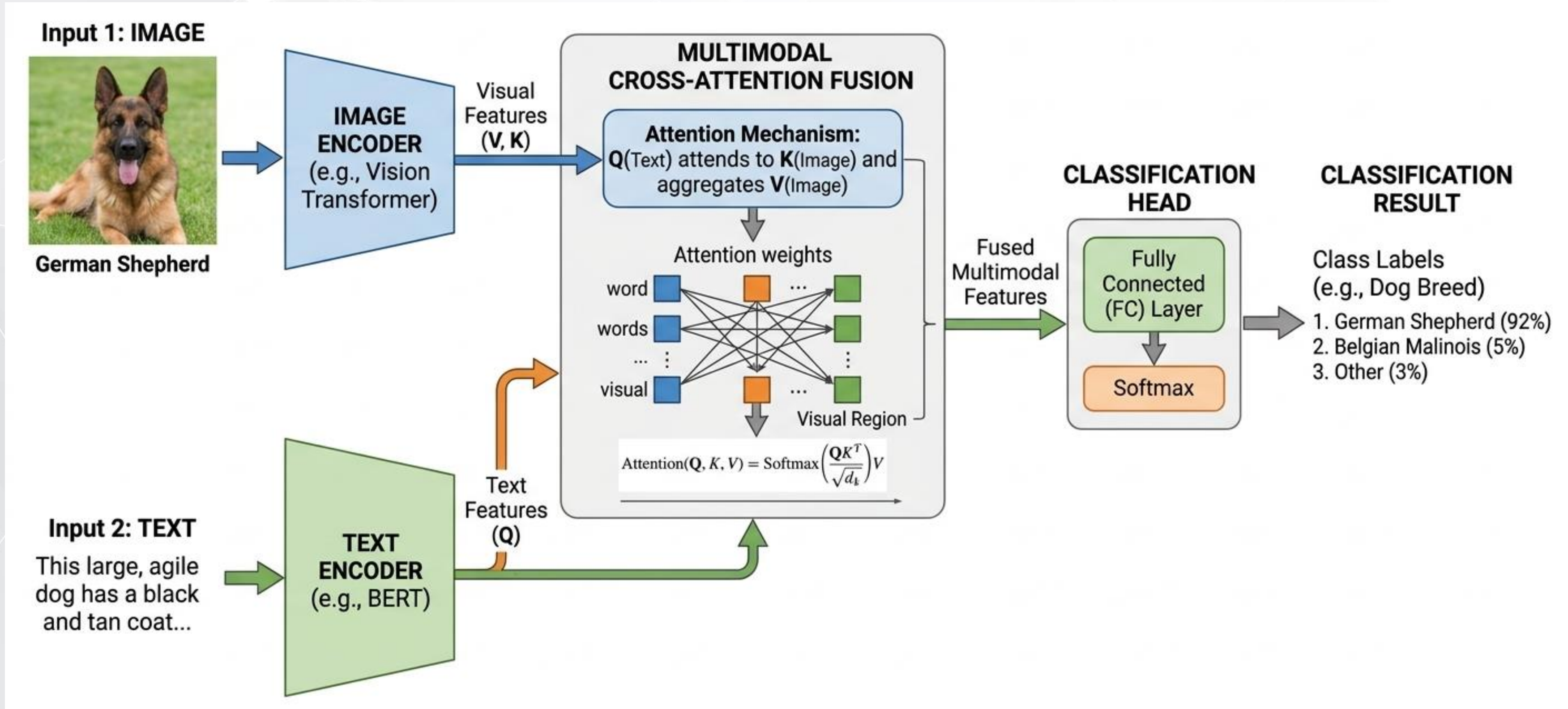
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Cross-Attention



Cross-Attention

